

ECO 310 - Problem Set 2

Question 5(c) - Full Solution

Problem: Assume that $u(w) = 1 - e^{-\alpha w}$, where $\alpha > 0$ (CARA utility function). Solve explicitly for the optimal insurance level q^* , and show that it does not depend on wealth w_0 .

Solution:

Step 1: Find the marginal utility

Given $u(w) = 1 - e^{-\alpha w}$, we have:

$$u'(w) = \alpha e^{-\alpha w} \quad (1)$$

Step 2: Apply the first-order condition from part (b)

From part (b), the optimal insurance level satisfies:

$$\frac{u'(w_0 - L + (1-p)q)}{u'(w_0 - pq)} = \frac{p(1-\pi)}{\pi(1-p)} \quad (2)$$

Step 3: Substitute the CARA marginal utility

Substituting $u'(w) = \alpha e^{-\alpha w}$ into both numerator and denominator:

$$\frac{\alpha e^{-\alpha(w_0 - L + (1-p)q)}}{\alpha e^{-\alpha(w_0 - pq)}} = \frac{p(1-\pi)}{\pi(1-p)} \quad (3)$$

The α terms cancel:

$$\frac{e^{-\alpha(w_0 - L + (1-p)q)}}{e^{-\alpha(w_0 - pq)}} = \frac{p(1-\pi)}{\pi(1-p)} \quad (4)$$

Step 4: Simplify using properties of exponents

Using the rule $\frac{e^a}{e^b} = e^{a-b}$:

$$e^{-\alpha(w_0 - L + (1-p)q) - (-\alpha(w_0 - pq))} = \frac{p(1-\pi)}{\pi(1-p)} \quad (5)$$

Expand the exponent:

$$-\alpha(w_0 - L + (1-p)q) + \alpha(w_0 - pq) \quad (6)$$

$$= -\alpha w_0 + \alpha L - \alpha(1-p)q + \alpha w_0 - \alpha pq \quad (7)$$

$$= \alpha L - \alpha(1-p)q - \alpha pq \quad (8)$$

$$= \alpha L - \alpha q[(1-p) + p] \quad (9)$$

$$= \alpha L - \alpha q \quad (10)$$

Notice that w_0 **cancels out completely!**

Therefore:

$$e^{\alpha L - \alpha q} = \frac{p(1-\pi)}{\pi(1-p)} \quad (11)$$

Step 5: Solve for q^*

Taking the natural logarithm of both sides:

$$\alpha L - \alpha q = \ln \left(\frac{p(1 - \pi)}{\pi(1 - p)} \right) \quad (12)$$

Solving for q :

$$\alpha q = \alpha L - \ln \left(\frac{p(1 - \pi)}{\pi(1 - p)} \right) \quad (13)$$

$$\boxed{q^* = L - \frac{1}{\alpha} \ln \left(\frac{p(1 - \pi)}{\pi(1 - p)} \right)} \quad (14)$$

Conclusion:

The optimal insurance level q^* depends only on the loss amount L , the risk aversion parameter α , the probability of an accident π , and the insurance price p .

Crucially, q^* **does not depend on initial wealth** w_0 , which is the defining property of CARA (Constant Absolute Risk Aversion) preferences. The demand for insurance is independent of wealth level.